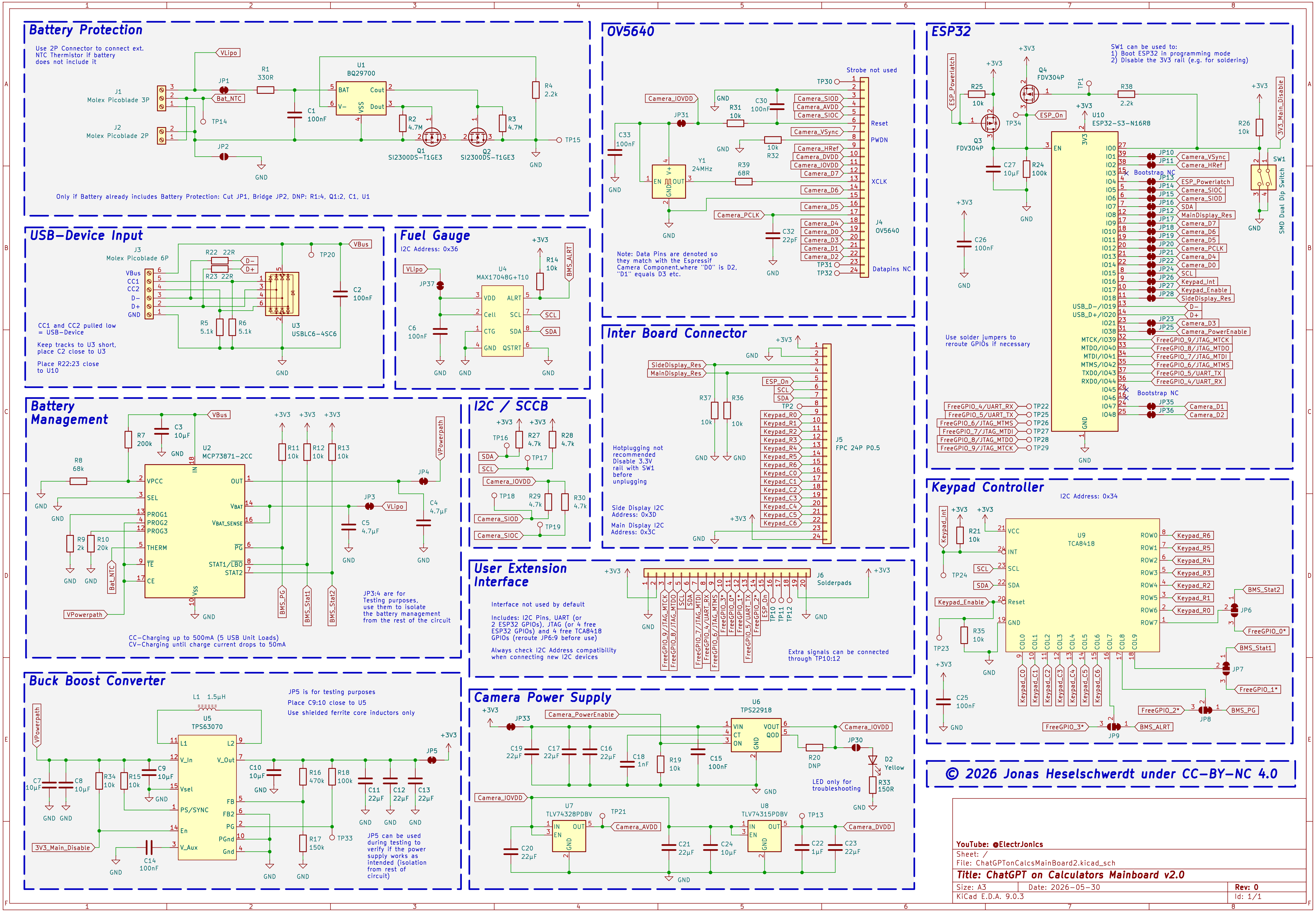
Battery Protection
Use 2P Connector to connect ext. NTC Thermistor if battery does not include it
J1 Molex Picoblade 3P
J2 Molex Picoblade 2P
VLipo
Bat_NTC
JP1
JP2
R1 330R
U1 BQ29700
BAT
V-
VSS
Cout
Dout
C1 100nF
R2 4.7M
R3 4.7M
R4 2.2k
Q1 SI2300DS-T1GE3
Q2 SI2300DS-T1GE3
TP14
TP15
GND
Only if Battery already includes Battery Protection: Cut JP1, Bridge JP2, DNP: R1:4, Q1:2, C1, U1
OV5640
Strobe not used
Reset
PWDN
XCLK
Datapins NC
J4 OV5640
Camera_IOVDD
Camera_SIOD
Camera_AVDD
Camera_SIOC
Camera_VSync
Camera_HRef
Camera_DVDD
Camera_D7
Camera_D6
Camera_D5
Camera_PCLK
Camera_D4
Camera_D0
Camera_D3
Camera_D1
Camera_D2
TP30
TP31
TP32
JP31
C30 100nF
C33 100nF
C32 22pF
R31 10k
R32 10k
R39 68R
Y1 24MHz
Note: Data Pins are denoted so they match with the Espressif Camera Component, where "D0" is D2, "D1" equals D3 etc.
ESP32
SW1 can be used to:
1) Boot ESP32 in programming mode
2) Disable the 3V3 rail (e.g. for soldering)
ESP_Powerlatch
R25 10k
Q3 FDV304P
Q4 FDV304P
TP1
TP34
ESP_On
R38 2.2k
R26 10k
3V3_Main_Disable
SW1
SMD Dual Dip Switch
U10 ESP32-S3-N16R8
C27 10µF
R24 100k
C26 100nF
Bootstrap NC
Use solder jumpers to reroute GPIOs if necessary
EN
IO0 IO1 IO2 IO3 IO4 IO5 IO6 IO7 IO8 IO9 IO10 IO11 IO12 IO13 IO14 IO15 IO16 IO17 IO18
USB_D-/IO19
USB_D+/IO20
IO21
IO38
MTCK/IO39
MTDO/IO40
MTDI/IO41
MTMS/IO42
TXD0/IO43
RXD0/IO44
IO45 IO46 IO47 IO48
Camera_VSync
Camera_HRef
ESP_Powerlatch
Camera_SIOC
Camera_SIOD
SDA
MainDisplay_Res
Camera_D7
Camera_D6
Camera_D5
Camera_PCLK
Camera_D4
Camera_D0
SCL
Keypad_Int
Keypad_Enable
SideDisplay_Res
D-
D+
Camera_D3
Camera_PowerEnable
FreeGPIO_9/JTAG_MTCK
FreeGPIO_8/JTAG_MTDO
FreeGPIO_7/JTAG_MTDI
FreeGPIO_6/JTAG_MTMS
FreeGPIO_5/UART_TX
FreeGPIO_4/UART_RX
Camera_D1
Camera_D2
TP22 TP25 TP26 TP27 TP28 TP29
USB-Device Input
J3 Molex Picoblade 6P
VBus CC1 CC2 D- D+ GND
VBus
R22 22R
R23 22R
TP20
R5 5.1k
R6 5.1k
U3 USBLC6-4SC6
C2 100nF
CC1 and CC2 pulled low = USB-Device
Keep tracks to U3 short, place C2 close to U3
Place R22:23 close to U10
Fuel Gauge
I2C Address: 0x36
U4 MAX17048G+T10
VDD ALRT Cell SCL CTG SDA GND QSTRT
JP37
C6 100nF
R14 10k
BMS_ALRT
Inter Board Connector
J5 FPC 24P P0.5
SideDisplay_Res
MainDisplay_Res
ESP_On
Keypad_R0 Keypad_R1 Keypad_R2 Keypad_R3 Keypad_R4 Keypad_R5 Keypad_R6
Keypad_C0 Keypad_C1 Keypad_C2 Keypad_C3 Keypad_C4 Keypad_C5 Keypad_C6
TP2
R37 10k
R36 10k
Hotplugging not recommended Disable 3.3V rail with SW1 before unplugging
Side Display I2C Address: 0x3D
Main Display I2C Address: 0x3C
Battery Management
U2 MCP73871-2CC
VPCC SEL PROG1 PROG2 PROG3 THERM TE CE Vss IN OUT VBAT VBAT_SENSE PG STAT1/LBO STAT2
R7 200k
C3 10µF
R8 68k
R9 2k
R10 20k
R11 10k
R12 10k
R13 10k
C4 4.7µF
C5 4.7µF
JP3
JP4
VPowerpath
BMS_PG
BMS_Stat1
BMS_Stat2
Bat_NTC
JP3:4 are for Testing purposes, use them to isolate the battery management from the rest of the circuit
CC-Charging up to 500mA (5 USB Unit Loads)
CV-Charging until charge current drops to 50mA
I2C / SCCB
TP16 TP17 TP18 TP19
R27 4.7k
R28 4.7k
R29 4.7k
R30 4.7k
Camera_IOVDD
Camera_SIOD
Camera_SIOC
User Extension Interface
J6 Solderpads
Interface not used by default
Includes: I2C Pins, UART (or 2 ESP32 GPIOs), JTAG (or 4 free ESP32 GPIOs) and 4 free TCA8418 GPIOs (reroute JP6:9 before use)
Always check I2C Address compatibility when connecting new I2C devices
FreeGPIO_0* FreeGPIO_1* FreeGPIO_2* FreeGPIO_3*
TP10 TP11 TP12
Extra signals can be connected through TP10:12
Keypad Controller
I2C Address: 0x34
U9 TCA8418
VCC INT SCL SDA Reset GND
ROW0 ROW1 ROW2 ROW3 ROW4 ROW5 ROW6 ROW7
COL0 COL1 COL2 COL3 COL4 COL5 COL6 COL7 COL8 COL9
R21 10k
R35 10k
C25 100nF
TP23 TP24
JP6 JP7 JP8 JP9
BMS_Stat2
BMS_Stat1
BMS_PG
BMS_ALRT
Buck Boost Converter
L1 1.5µH
U5 TPS63070
L1 L2 V_In V_Out Vsel PS/SYNC FB FB2 PG En PGnd V_Aux Gnd
JP5 is for testing purposes
Place C9:10 close to U5
Use shielded ferrite core inductors only
C7 10µF
C8 10µF
C9 10µF
C10 10µF
C11 22µF
C12 22µF
C13 22µF
C14 100nF
R34 10k
R15 10k
R16 470k
R17 150k
R18 100k
TP33
JP5
3V3_Main_Disable
JP5 can be used during testing to verify if the power supply works as intended (isolation from rest of circuit)
Camera Power Supply
U6 TPS22918
VIN CT ON VOUT QOD
JP33
Camera_PowerEnable
C19 22µF
C17 22µF
C16 22µF
C18 1nF
R19 10k
C15 100nF
R20 DNP
JP30
D2 Yellow
R33 150R
LED only for troubleshooting
U7 TLV74328PDBV
U8 TLV74315PDBV
TP21 TP13
Camera_AVDD
Camera_DVDD
C20 22µF
C21 22µF
C24 10µF
C22 1µF
C23 22µF
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Size: A3
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